

# Towards Bridging Expertise Gaps Through Human-Centered XAI in Clinical Settings

JOSH SHELL, University of Cincinnati, USA

ASHLEY KUENZI-DAVIS, Cincinnati Children's Hospital Medical Center, USA

JUN BAI, University of Cincinnati, USA

JILLIAN AURISANO, University of Cincinnati, USA

AI mediation in healthcare is often framed around the patient-clinician relationship. However, many high-stakes decisions occur upstream, where multiple stakeholders must interpret model outputs and decide whether they are trustworthy and how they should be used. In these settings, no single person holds the full context needed to determine whether a result is meaningful or biologically plausible. We ground this position in ongoing work on explainable AI (XAI) in cell-type classification for single-cell RNA sequencing (scRNA-seq) data. This work involves three roles: the machine learning practitioner who builds the model, the cellular biologist who evaluates the data, and the downstream researcher who depends on the annotations. Our observations suggest that explanation alone is insufficient. Instead, these workflows require systems that support interpretation and communication across roles with partial, non-overlapping expertise. We use our prototype and early findings to motivate this perspective and outline implications for AI-supported collaboration in healthcare.

Additional Key Words and Phrases: Explainable AI, Healthcare, User-Centered Design

## 1 Introduction

AI models are becoming increasingly capable at biomedical classification tasks. In single-cell RNA sequencing (scRNA-seq), automated cell-type annotation has been benchmarked across countless datasets, and strong performance metrics are routinely achievable[4]. But accuracy is not the same as plausibility, and a model that performs well does not automatically produce outputs that stakeholders can evaluate, communicate, or understand. What happens *after* the model works is the harder problem. Making use of these predictions requires negotiation and shared decision-making across roles with non-overlapping expertise: the AI engineer who built the model, the biologist who understands the data and must approve the results, and the downstream user who depends on the annotations. Bridging these roles requires more than a single explanation interface. It requires AI systems that can support two distinct functions: providing explainability metrics at the feature level to illuminate model behavior, and helping stakeholders assess whether those features align with biological understanding of what distinguishes cell types. This work sits at that intersection: a design probe in XAI for scRNA-seq cell-type classification that exposes multi-stakeholder dynamics following the nested model approach [5].

## 2 Limits of Single-User XAI Framings

Much of XAI assumes the user is seeking to understand how a model arrived at a prediction [3]. This often assumes explanations can be evaluated against known ground truth. However, in biomedical settings ground truth is often incomplete or contested. In scRNA-seq classification, labels may be uncertain, marker genes vary across species and

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Authors' Contact Information: Josh Shell, University of Cincinnati, USA, ShellJl@mail.uc.edu; Ashley Kuenzi-Davis, Cincinnati Children's Hospital Medical Center, USA, Ashley.Kuenzi@cchmc.org; Jun Bai, University of Cincinnati, USA, baiju@ucmail.uc.edu; Jillian Aurisano, University of Cincinnati, USA, aurisajm@ucmail.uc.edu.

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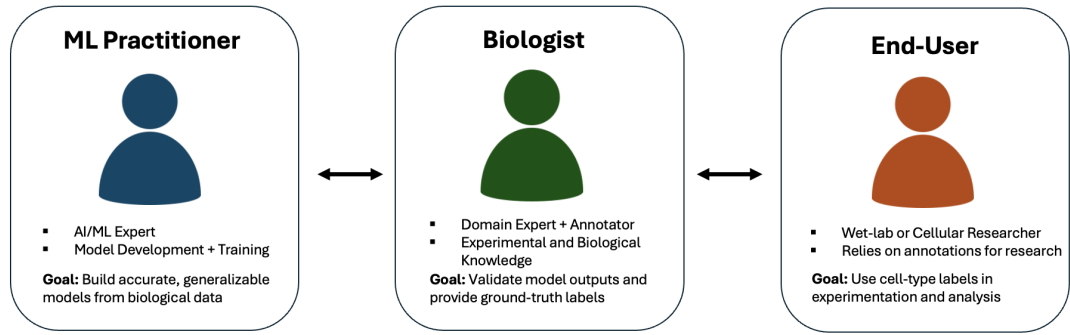


Fig. 1. Three stakeholder roles in automated scRNA-seq classification: ML practitioner, biologist, and downstream researcher. Each brings partial expertise with minimal overlap, requiring negotiation to interpret model outputs

tissue types, and experts must reconcile model outputs against their own evolving background knowledge. The question becomes not whether a prediction is correct, but whether it is biologically plausible.

### 3 The Stakeholder Triangle

Through our work we have identified three core roles in interpreting AI model outputs in this domain: **The ML practitioner** understands architecture, parameters, training decisions, and evaluation metrics but lacks biological context. **The biologist** understands gene expression, scRNA-seq workflows, and data quality but lacks advanced AI knowledge. **The downstream researcher or client** relies on those annotations, primarily concerned with correctness. This structure highlights why explanation alone is insufficient. Each role asks different questions of the same output, and meaningful interpretation emerges through interaction, negotiation, and comparison across these perspectives[6].

### 4 Design Probe: scRNA-seq Classification

The stakeholders must negotiate decisions from different vantage points: the ML practitioner knows what the model selected and why, the biologist knows whether those features make biological sense, and the downstream researcher needs to trust the result. No single interface addresses all three, so the question becomes what shared representations can support that negotiation. We used ML-predicted cell-type classification as a design probe to study how stakeholders reason under uncertainty. Our prototype grounds familiar workflows in model-derived XAI metrics: a UMAP for cluster-level reasoning, XAI panels showing instance-level feature importance alongside class-level weights, and dot plots comparing user and model selected genes. We also included a language model component to act as an aid, translating between ML and biological explanations of the same features so that stakeholders can question the model's reasoning together rather than in isolation.

### 5 Position and Implications

AI already exists in pre-clinical workflows, particularly in scRNA-seq classification, and its role will only grow as datasets and computation power scale. For AI practitioners, model accuracy does not address if the outputs are biologically plausible or fit their downstream use. Traditional XAI, as currently framed is not sufficient because it addresses one user's understanding of the problem. Our gap is different: supporting explanation and negotiation across stakeholders

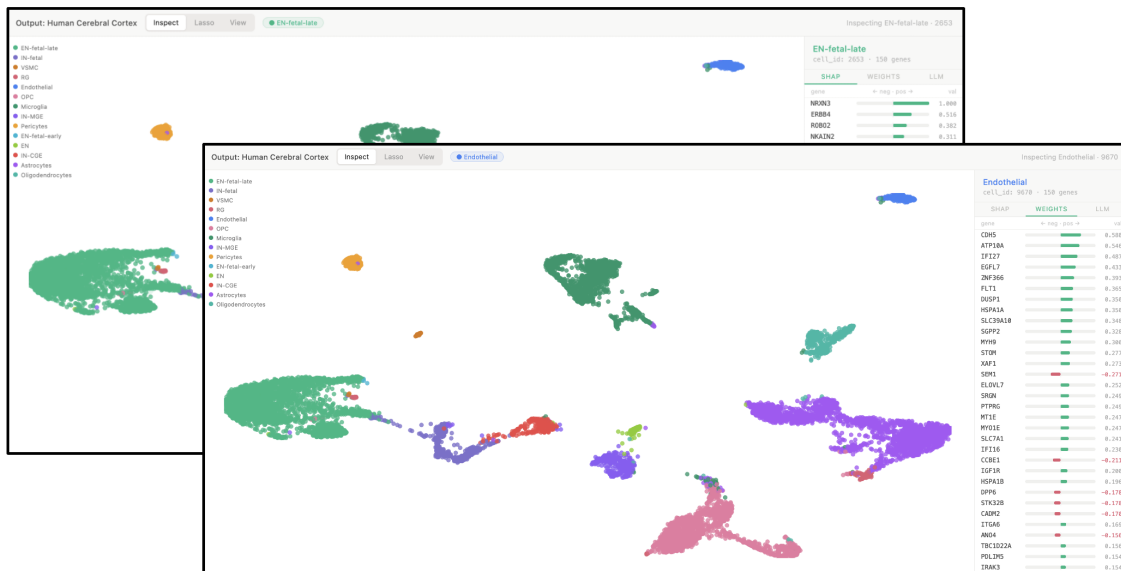


Fig. 2. Prototype interface showing cluster-level UMAP projections. Selecting a cell reveals feature-level XAI metrics to support interpretation across stakeholders

with partial, non-overlapping knowledge. This reveals three implications for the design of AI systems in this space, building on established interaction guidelines[1]:

**Role-aware XAI** systems should adapt to different stakeholders, not one-fits-all interface. **Multi-directional explanations** must support communication across roles as AI experts, biologists, and end-users face different translation problems. **Reasoning under uncertainty** systems should present evidence and enable comparison rather than asserting certainty. When ground truth is unknown or evolving, the goal is supporting plausibility, not asserting truth. We offer this as an invitation to expand where and how XAI in healthcare is applied. The pre-clinical layer, where data is processed, labeled, and interpreted, directly shapes downstream clinical decisions. Here, the problem is not explaining a model to a single user, but supporting how multiple stakeholders interpret, question, and communicate model outputs together [2]. If we want AI to support real-world decision making in healthcare, we need to design for this layer explicitly, not just accuracy.

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